

RPG Creator

One-page app summary from repo evidence

What it is

A C++ desktop app for building RPG-style maps with a separate TCP server for shared projects. The repo shows pixel and sprite editing, project sharing, chat, and export/import flows.

Who it is for

Primary persona: RPG or game-map creators collaborating on shared 2D map projects.

What it does

- Authenticates users and loads their project list.
- Creates, renames, duplicates, deletes, shares, and joins projects.
- Edits maps with pixel layers and sprite layers.
- Draws and erases square, circle, and diamond brush shapes.
- Places, moves, resizes, rotates, and removes sprites.
- Manages members and roles: spectator, editor, owner.
- Supports chat plus PNG and native .nif export/import.

How it works

- Client: SFML + TGUI desktop UI with Qt-based data handling. ``Application`` swaps ``LoginView``, ``MenuView``, and ``GameView``, while ``Project``, ``Map``, ``ToolBar``, and ``Chat`` hold editor state.
- Transport: ``ClientNetworkManager`` sends protocol messages over TCP to ``127.0.0.1:5001``; ``ServerNetworkManager`` accepts sockets, queues requests, and sends responses.
- Server/data: ``Worker`` mutates in-memory ``LiveProject`` sessions and ``LayerManager``; ``SaveWorker`` persists project JSON, images, zipped exports, and chat via ``ProjectsManager``, while ``DatabaseManager`` stores users, projects, roles, and share tokens in SQLite ``db.sqlite``.

How to run

- From repo root: ``chmod +x install.sh`` then ``./install.sh``.
- The documented install path is apt-based Linux; ``install.sh`` uses ``sudo apt`` and builds into ``bin/``.
- Start the server from ``bin/``: ``./server``.
- Start the client from ``bin/`` in another terminal: ``./client``.